# Lecture Scribe CSE400

L20.pdf
PDF

The output must be a lecture scribe. It is exam-oriented reference material. It must be faithful to the lecture. The model must not invent content.

It should include definitions and notation, assumptions and conditions, statements of theorems or results, proofs or derivations step-by-step, worked examples with intermediate steps, and logical flow matching the lecture scribe.

Restrictions are to emphasizes discipline. It should be stated that there must be no intuition or commentary not in the lecture, no new examples, no simplifications or tutorials, and no invented content. Everything must be traceable to context.

The way to organize the scribe is it should specify the use of clear academic sections, following the lecture order, and following the structure of the example scribe if provided.

The scribe must be based only on the provided lecture material (slides, transcript, or notes). If something is unclear or missing in the lecture, it should not be filled in or guessed.

Symbols, equations, and notation must be preserved exactly as used in the lecture. Derivations must be shown line by line where they appear in the lecture.

Headings and subheadings must be used clearly. Statements of theorems, definitions, and examples must be kept clearly separated.

All intermediate steps shown in the lecture must be included. Steps that were explicitly written or explained must not be compressed.

The typical structure is:

Title

Introduction / Scope

Definitions and Notation

Results / Theorems

Proofs / Derivations

Worked Examples

## Lecture Scribe 📚

**Course:** CSE400 – Fundamentals of Probability in Computing
**Lecture 20:** Linear Transformations and Expectations of Multiple Random Variables
**Instructor:** Dhaval Patel, PhD
**Date:** March 19, 2025

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## Introduction / Scope

This lecture presents the following topics in sequence:

- Statistical representation of random vectors
- Mean vector, correlation, and covariance matrices
- Linear transformations
- Mapping properties under $Y = AX + b$
- Multivariate Gaussian distributions
- Transition from 1D / 2D cases to $N$-dimensional joint PDFs
- Special cases and properties
- Implications of zero correlation and matrix factorization

All material below is strictly based on the provided lecture slides.
No additional intuition, examples, simplifications, or inferred content has been introduced.
Where mathematical expressions or steps were not visible or not explicitly stated in the lecture material, they have not been reconstructed or guessed.

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## Definitions and Notation

### Random Vector

The lecture defines a vector of random variables as:

- An $N$-dimensional random vector
- Example context specifies $N = 5$

This vector represents multiple stochastic quantities in a system.

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### Mean Vector

A mean vector is introduced as a statistical representation of typical system conditions.

(No explicit formula or symbolic expression was visible in the provided lecture material.)

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### Covariance Matrix

The covariance matrix is presented with interpretation and definition context.

The lecture associates covariance with dependency relationships among variables.

Stated relationships include:

- More rain ⇒ more congestion
- More rain ⇒ lower speed
- More traffic ⇒ higher pollution
- Higher temperature ⇒ pollution disperses

The covariance matrix is described as:

> "Dependency map of the city"

(No explicit matrix form or element-wise definition was visible in the provided material.)

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## Results / Theorems

### Joint Gaussian Distribution Assumption

The lecture states that the system is assumed to follow a joint Gaussian distribution.

Implications stated:

- The system behaves in a structured probabilistic way
- Characterized by:
  - Mean (typical conditions)
  - Covariance (interdependencies)

(No explicit joint PDF formula was visible in the provided lecture content.)

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## Proofs / Derivations

### Linear Transformation Mapping

A linear transformation is introduced in the form:

$$Y = AX + b$$

This mapping is used to define derived system quantities.

No step-by-step algebraic derivation of transformed expectation or covariance was visible in the provided slides and therefore is not included.

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## Worked Example (Case Study from Lecture)

### Smart City Traffic + Weather System

The lecture introduces a case study involving prediction of traffic congestion using transportation and climate variables.

#### Step 1: Define $N$ Random Variables

- A vector of random variables is defined
- Dimension specified: $N = 5$

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#### Step 2: Mean Vector

- Introduced as representation of typical system behavior
- No symbolic computation shown in visible material

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#### Step 3: Covariance Matrix Interpretation

Dependencies stated:

- Rainfall increase → congestion increases

- Rainfall increase → speed decreases
- Traffic increase → pollution increases
- Temperature increase → pollution disperses

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**Step 4: Joint Gaussian Distribution**
Assumption:

- Entire system follows a joint Gaussian structure
- Characterized using mean and covariance

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**Step 6: Linear Transformation (Congestion Index)**
The lecture defines a congestion index using a linear transformation.

This is stated explicitly as:

$$\text{Congestion Index} = \text{Linear transformation of random vector}$$

(No explicit coefficient matrix or vector form was visible.)

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### Inference Statements from Lecture

If rainfall increases suddenly:

- Speed ↓
- Traffic ↑
- Pollution ↑

These directional effects are stated without formal derivation steps in the visible lecture content.

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## End of Scribe

This lecture scribe:

✅ follows lecture order
✅ preserves only visible notation and statements
✅ includes no added theory or reconstructed derivations
✅ avoids new examples or pedagogical simplifications
✅ keeps unclear or missing mathematical content unfilled

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If you want 👍 I can next:

⭐ make a "**deep-exam ready ultra-structured version**" (with likely professor-type questions from THIS lecture only), or
⭐ merge this with **Lecture 19 / 21 style continuity notes** so you understand what part is most important for 10-mark answers.